

Ke Shi

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EDUCATION

University of Electronic Science and Technology of China <i>B.S. in Computer Science (Honors); GPA: 3.91/4.00</i>	Chengdu, China <i>Sep 2019 – Jun 2023</i>
University of Science and Technology of China <i>Ph.D. student in Computer Science; Advisor: Shuai Shao; GPA: 4.11/4.30</i>	Hefei, China <i>Sep 2023 – Jun 2028 (Expected)</i>

RESEARCH DIRECTIONS

Algorithms, combinatorial optimization, approximate counting, probabilistic combinatorics, and connections to phase transitions in statistical physics.

PUBLICATIONS

Zero-Freeness of the Hard-Core Model with Bounded Connective Constant <i>Yuan Chen, Shuai Shao, Ke Shi</i>	<i>ISAAC 2026</i>
Strong Spatial Mixing for Continuous Spin Systems with Multi-Interactions <i>Shuai Shao, Ke Shi, Bohan Wang</i>	<i>ISAAC 2026</i>
Zero-Freeness is All You Need: A Weitz-Type FPTAS for the Entire Lee–Yang Zero-Free Region <i>Shuai Shao, Ke Shi</i>	<i>ITCS 2026</i>
An Optimal Algorithm for the Stack Crane Problem on Fixed Topologies <i>Yike Chen, Ke Shi, Chao Xu</i>	<i>ISAAC 2025</i>
Almost optimum ℓ-covering of $\mathbb{Z}_n$ <i>Ke Shi, Chao Xu</i>	<i>COCOON 2024</i>
A Polynomial Time Algorithm for Finding a Minimum 4-Partition of a Submodular Function <i>Tsuyoshi Hirayama, Yuhao Liu, Kazuhisa Makino, Ke Shi, Chao Xu</i>	<i>SODA 2023 & Mathematical Programming 2023</i>

OPEN-SOURCE CONTRIBUTIONS

Google Summer of Code 2026 <i>Contributor, exact HMC for volume-based counting of linear extensions</i>	GeomScale/volesti <i>May 2026 – Aug 2026</i>
Open Source Promotion Plan 2025 <i>Mentor, Emoji Support Project</i>	Mogan STEM <i>Jul 2025 – Sep 2025</i>
Google Summer of Code 2024 <i>Contributor, modernized Dingo's linear-programming solver interface</i>	GeomScale/dingo <i>Jun 2024 – Aug 2024</i>
Open Source Promotion Plan 2024 <i>Contributor, OpenType MATH support</i>	Mogan STEM <i>Jul 2024 – Sep 2024</i>

MANUSCRIPTS

Strong spatial mixing for the matching polynomial: A simple proof via Heilmann–Lieb Theorem <i>Shuai Shao, Ke Shi</i>	<i>2025, Submitted</i>
An LP Algorithm for Counting Eulerian Orientations Through the Lens of Quasi-polymorphism <i>Jincheng Guan, Shuai Shao, Ke Shi</i>	<i>STOC '26 “Can AI Do Theory?” workshop; 2026, submitted</i>
Coupling Independence implies Zero-Freeness <i>Shuai Shao, Ke Shi</i>	

SELECTED HONORS

National College Student Mathematics Competition <i>12th & 13th competitions: First Prize (Finals), 2021 & 2023</i>	Champion (Finals), 2023
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